



Renewable Energy Plant Development, Historic Environments, and Local Communities: Lessons from the Greek Experience

Ioannis Poulios

To cite this article: Ioannis Poulios (2012) Renewable Energy Plant Development, Historic Environments, and Local Communities: Lessons from the Greek Experience, The Historic Environment: Policy & Practice, 3:2, 127-142

To link to this article: <http://dx.doi.org/10.1179/1756750512Z.00000000011>



Published online: 12 Nov 2013.



Submit your article to this journal [↗](#)



Article views: 20



View related articles [↗](#)



Citing articles: 2 View citing articles [↗](#)

Renewable Energy Plant Development, Historic Environments, and Local Communities: Lessons from the Greek Experience

IOANNIS POULIOS

Hellenic Open University, Greece

The development of renewable energy plants (REPs) is mostly seen from the technical, economic, and legal perspectives, with the cultural and social aspects often not being taken into account, sometimes resulting in a serious impact on the progress of the investment. This is particularly evident with historic environments, where local communities tend to have a negative attitude towards the development of REPs, causing severe delays and even the cancellation of the investment. This article examines the impact of REPs on historic environments in relation to the attitude of local communities, using a variety of case studies from Greece. The aim is to develop a new approach of broader applicability that would link the development of REPs to the protection of historic environments and the sustainable development of local communities, and then provide specific recommendations (consisting of practical steps) for the implementation of this approach to how a renewable energy project can succeed while at the same time respecting the historic environment and engaging the local community.

KEYWORDS renewable energy plant, historic environment, local community, sustainable development, Greece

The Greek context

The development of renewable energy plants (henceforth cited as REP) is high on the political agenda of the Greek government, in an attempt to comply with the European Union (EU) regulations and also as a potential outlet from the current financial crisis. As the former Prime Minister of Greece stated: ‘a planned long-term policy for the promotion of the Renewable Energy Sources and for the gradual detachment from oil and conventional fuels, in parallel with the creation of new business opportunities

and employment positions, consists a strategic objective'.¹ REPs, by law, are considered projects of 'public benefit'.²

The development of REPs, as with any type of urban development project, has to ensure respect for the historic (as well as the natural) environment, and presupposes the conduct and approval of an Environmental Impact Assessment, in the context of 'integrated' and 'holistic' conservation.³ Respect for the historic environment in Greece is controlled by a strict legislative framework: the ownership of the environment is in the hands of the state, while the protection of the environment is recognized as 'the duty of the state and the right of everyone'.⁴ The system of heritage protection is particularly concerned about 'the safeguarding of [the monuments]' material substance and their authenticity'⁵ in an attempt to 'safeguard the historic memory for the sake of the present and the future generations',⁶ and could be generally characterized by an emphasis on the preservation of the tangible/material — rather than the intangible — elements of monuments.⁷ The responsibility for protection lies with the Ministry of Culture; even in those cases where other government agencies (such as the Ministry of Environment, Energy, and Climate Change) are also involved, the Ministry of Culture retains final responsibility. Protection is centrally administered: the central offices of the Ministry set the policy, while the local offices deliver this policy at the local level.⁸

On a practical level, respect for the historic environment and, more specifically, the conduct and approval of an Environmental Impact Assessment faces a number of difficulties. The criteria that comprise the 'material substance and authenticity' (see above) remain unclear, and guidelines on ways to define and safeguard the historic memory and take into account and involve the present generation, let alone future ones, are lacking. There is no complete and updated national archive of listed monuments, and a large number of listed monuments remain unrecorded to date. The responsibilities of the differing government bodies involved are not well defined, and cooperation between the investor with the central and local government offices in most cases is far from satisfactory. Also, there are no specific and detailed requirements for the conduct and approval of an Environmental Impact Assessment.⁹

Despite this difficult context for the development of REPs, certain investments succeed in moving on to the stage of implementation, only to face a usually negative attitude on the part of the local community, often resulting in considerable delays and even the cancellation of the investment. Therefore, the development of REPs in Greece is placed — as far as protection of the historic environment is concerned — under the same legislative framework as any other type of development project: a strict and government-controlled framework that focuses on the preservation of the material of the monuments. In practice, however, the development of REPs seems to present further complexities that may differentiate them from an average development project:

- the significant economic potential of REPs, coupled with the political pressure for their development, tends to give a harder push on the part of the investors
- the rather insufficient experience of the government authorities (especially the officials of the Ministry of Culture) in the development of REPs — given that REPs are a relatively new type of development — leads to vague and confusing guidelines for the protection of the historic environment and to a particularly



FIGURE 1 Map of Greece showing the locations of the case studies of the historic environments discussed in this article.

SOURCE: Google Earth™

difficult cooperation with investors, while political pressure for the development of REPs tends to place the government authorities (especially the officials of the Ministry of Culture) in an even more inconvenient position, and

- the scale of REPs and their potential impact on the historic environment seem to make the local communities move into a cautious and even reluctant stance.

Examples from Greece: Presentation

The Acheloos River Diversion Scheme (close to St George Monastery at Myrofyllo, Prefecture of Trikala)

The scheme for the diversion of the Acheloos River to the valley of Thessaly in central Greece (Figure 1) calls for the completion of a number of projects: a major diversion channel, two tunnels, a water intake system, and a hydroelectric power station with a series of dams being built by the Public Power Corporation of Greece. The scheme as a whole cannot be considered a Renewable Energy project, but part of it is related to Renewable Energy.

Significance of the historic environment (with an emphasis on the association of the local community with the historic environment)

St George Monastery at Myrofyllo (Figure 2) is a seventeenth-century monastery that is no longer living — in the sense that it does not currently host a monastic



FIGURE 2 St George Monastery at Myrofylo.

By permission of Giorgos Chondros

community — but is operated by the inhabitants of the nearby villages on certain occasions every year (e.g. on the feast day of the patron saint), and also has a symbolic value given the role it played for the area in the Greek Revolution (1821–32).

REP development process (with an emphasis on the attitude of the local community)

The Environmental Impact Assessment for the project referred to the Monastery, but considered its archaeological value not to be significant and did not take into account its symbolic value. Thus, the flooding of the monastery, as a result of the construction of the hydroelectric power station, was proposed as unavoidable for the implementation of the project. The Ministry of Culture and all the government bodies involved gave their consent because the project was — at that time — considered vital for the development of the region of Thessaly, and the hydroelectric power station was about to begin construction. However, after the intervention of environmental non-governmental organizations (NGOs) and then the inhabitants of the nearby villages, the Council of State (the highest court in Greece) decided that alternative options had to be considered that would save the Monastery. Therefore, the initial plans had to change, with the proposal for the construction of a protective wall around the monastery, leading to major delays in the progress of the project.¹⁰ The Acheloos River Diversion Scheme was restrained again — for different reasons — by the Council of State, and has now been finally stopped.

The operation of a wind farm near Hosios Meletios Monastery, Prefecture of Boiotia

Significance of the historic environment (with an emphasis on the association of the local community with the historic environment)

Hosios Meletios Monastery dates back to the twelfth century and is associated with the reformation of monastic life in Greece. Today it continues to function as a living monastery.

REP development process (with an emphasis on the attitude of the local community)

A proposal for the construction of a wind farm in an area near the Monastery was approved by the Ministry of Culture on the grounds that its distance from the monument was considered sufficient and thus there would be no considerable optical interference to the monument, and the wind farm was completed. However, the monastic community argued that the noise generated by the wind turbines hindered them from the conduct of their religious rituals and their everyday-life duties, and called for the cessation of the operation of the wind farm. The view of the monastic community was supported by an NGO. The Council of State decided on the cessation of a number of the wind turbines, so that the noise generated would be decreased.¹¹

Proposals for the construction of wind farms on Gyaros Island, Prefecture of Cyclades

Significance of the historic environment (with an emphasis on the association of the local community with the historic environment)

Gyaros is a very small, arid, and unpopulated island in the Aegean Sea. The island was used as a place of exile and torture for political dissidents in Greece in various periods between 1948 and 1974, during the Civil War and the military junta, with the remains of the main prison building and camps still visible today. The island as a whole is recognized as a place of historic memory associated with the struggles of the Greek people for freedom and democracy. The symbolic value of the island is vivid and strong, particularly for former prisoners, who have established an association for the safeguarding of the historic memory of the place, 'Gyaros Historical Memory', and once a year pay tribute to their departed colleagues by holding ceremonies at the cemetery on the island.

REP development process (with an emphasis on the attitude of the local community)

For the exploitation of the rich wind energy potential of the island, two proposals were submitted (by two different firms) for the construction of wind farms on the part of the island where there were no remaining structures. The proposals met the silent consent of the Municipality of Syros, to which the island belongs in terms of administration, and the population of the nearby islands, but the firm opposition of the 'Gyaros Historical Memory' on the basis of the potential impact of the investment on the historic memory of the island. The issue is still in debate.¹²

The construction of a wind farm on Naxos Island, Prefecture of Cyclades

REP development process (with an emphasis on the attitude of the local community)

A proposal for the construction of a wind farm on Naxos Island was approved by the Ministry of Culture, and the project was about to begin. Despite this approval,

transference of wind turbines to the area of interest was conducted without any prior study, causing the destruction of non-listed roads and bridges within the village of Chalkeios (Figure 3) and affecting the surrounding environment of the listed aerial railway of the island. The inhabitants of the island did not react, at least in an active way, while the Ministry of Culture missed the incident, and thus the project moved on.¹³ On the other hand, it is not certain under what circumstances and whether in the first place the project would have proceeded had any preliminary study been carried out, or had the inhabitants or the Ministry reacted to the damage caused by the transference of the wind turbines.

The revival of the traditional system for water management on Amorgos Island, Prefecture of Cyclades

While this is a heritage restoration rather than an energy investment project, it is included in this study because it helps with the development of certain aspects of the recommendations set out below.

Significance of the historic environment (with an emphasis on the association of the local community with the historic environment)

Amorgos Island's system for the collection and management of water resources, consisting of cisterns, fountains, and tanks, is associated with local practices and traditions; but the system was no longer in use and was in danger of collapsing.



FIGURE 3 Transferring wind turbines through the village of Chalkeios, Naxos Island.
By permission of Eirini Gratsia

REP development process (with an emphasis on the attitude of the local community)

An NGO, with the support of the Municipality of Amorgos and the Prefecture of Cyclades, succeeded in restoring and reviving the system as well as the associated practices and traditions. This was achieved by benefiting from local knowledge on the use of the system and engaging the community from the very beginning of the project.¹⁴

Examples from Greece: Review

The examples cited above demonstrate that the cultural and social aspects were not taken into account, at least sufficiently, in the planning process for the development of REPs, resulting in a serious impact on the progress of the investments. The core issue was that the planning process created a dichotomy between the historic environment and the local community, sensing and treating the local community as an external or supplementary aspect detached from the historic environment. This approach resulted in a series of mistakes throughout the process (mistakes that are present to some extent in the legislation, affect the attitude and the guidelines on the part of the government bodies and their cooperation with investors, and rule the practices of those investors).

First, REPs did not link the significance of the historic environment to the association of the local community with the historic environment. As a result, the significance of the historic environment was mostly sensed in terms of the material/tangible expressions and associated values, while the intangible expressions and associated values, to a considerable extent, were neglected. Indicative examples are as follows:

- in the case of St George Monastery, the religious value (through its use by the inhabitants of the nearby villages) and the symbolic value (given its role in the Greek Revolution) were not sufficiently considered
- in the case of Hosios Meletios Monastery, the historic value (of the monastery structures) and the aesthetic value (avoidance of the optical interference to the monastery) were emphasized, while the religious value (through the practices of the monastic community) and the symbolic value (as a place associated with the reformation of the monastic life in Greece) were not taken into account
- with regard to Gyaros Island, the historic value (of the remaining structures) was given priority over its symbolic value (through the associated memories and the rituals of the former prisoners).

Moreover, the boundaries of the historic environment in some cases were defined on the basis of material/tangible expressions rather than intangible ones. For example, the boundaries of Hosios Meletios Monastery were defined on the basis of the monastery structures (through avoidance of the optical interference to the structures) and not on monastic practices (through negligence of the acoustic disturbance to the practices). Similarly, in defining the boundaries of Gyaros Island, emphasis was on the physical structures (covering a specific part of the island), rather than the historic memory (covering the entire territory of the island).

Second, REPs sensed and approached all historic environments — as well as all local communities — in the same way, in a ‘one size fits all’ approach, regardless of

the differing ways in which the association of the local community with the historic environment had evolved over time in each case. For example, Hosios Meletios Monastery, which has continued to function as a living monastery operated by the associated monastic community, was treated in the same way as St George Monastery, which had lost its monastic function and was operated by the inhabitants of the nearby villages. In the case of Gyáros Island, the community of the former prisoners, who had a strong symbolic connection with the island and were actively involved in its operation and management, were considered equally to the inhabitants of the nearby islands.

Third, REPs did not see the historic environment, through its association with its local community, as being in continuous evolution over time and into the future, and in some cases appeared to focus on its current/present phase (the one ‘hosting’ the investment) as if it were the final one. As a result, attention was given to the REP’s main construction phase, rather than the pre- and especially the post-construction phase (i.e. the aftermath of the investment). The omission to carry out a pre-construction phase was illustrated in the case of Naxos Island, where no provision was made for transferring the elements of the wind turbines through the inhabited landscape. Failure in the post-construction phase was evident in the case of Hosios Meletios Monastery where no provision was made for the impact of the operation of the wind turbines on the practices of the monastic community, and also in the case of the Acheloos River Diversion Scheme, whose construction would have led to radical changes in the broader landscape and the economy and life of the local community.

Fourth, REPs did not involve the local community in the process. There was insufficient awareness of the characteristics of the local community — in terms of structure, management systems, processes and people, and concerns, — and there was no provision for the establishment of a formula of cooperation with the local community. The local community was taken into account only at a very late stage in the process, when it could not have been possibly avoided. At that stage, the local community retained a negative attitude towards the development of REP, causing severe delays and even the cancellation of the investments. The attitude of negligence towards the local community and the consequences on the progress of the investment were evident in the cases of St George Monastery, Hosios Meletios Monastery, and Gyáros Island.

Towards a new approach (of broader applicability)

A new approach is required for the development of REPs to link them to the protection of the historic environment and the sustainable development of the local community. This new approach is based on the notion that the local community is treated as an inseparable part of the historic environment, and follows certain key principles.

First, REPs must link the significance of the historic environment to the association of the local community with the environment. An historic environment, defined through its association with the local community, consists of the following elements:¹⁵

- a) The function of the historic environment. Indicative examples: religious environment, and traditional settlement.

- b) The material/tangible and the intangible expressions associated with the historic environment, which should be given equal attention. Material expressions include: the material type, construction technique, and decoration of a monument; the internal and external space of a monument; any structure that exists on the surface and underneath the historic environment. The intangible expressions include myths, habits, customs, traditions, beliefs, and rituals associated with an environment.
- c) The values inscribed to the historic environment by the community. Values include those associated to the material expressions, such as historic, aesthetic, scientific/research, and economic ones, and those associated to the intangible expressions, such as social, religious, and symbolic ones,¹⁶ which should be given equal attention.
- d) Community's care of the historic environment, expressed through the ownership status, the management systems, and the maintenance practices. These can be traditional ones (such as customary law, sacred and pragmatic controls, and traditions:¹⁷ for example, the Abbot and the Council of Hosios Meletios Monastery) or modern, scientific-based ones (such as modern legislation).

Second, REPs see each historic environment, through its association with its community, as a unique one, and treat them differently. Several types of historic environment may be identified, based on different types of community association with the environments, each association being of different strength. These different types have to do with the differing ways in which the community's association with the historic environment has evolved over time to date, in terms of function, material/tangible and intangible expressions, values, and care (see above). Specifically, the following types of historic environment may be identified (from the weakest to the strongest association of communities with the environment):¹⁸

- An historic environment with a (simply) local community — i.e. any community living in or near a historic environment — whose association with the environment is a relatively weak one. In the example of Gyaros Island (though, strictly speaking, there is no 'local' community in the sense that the island is uninhabited) the inhabitants of the nearby islands, which is the community with the closest geographical proximity to the island, have a very weak association with the historic environment of Gyaros, as expressed through their consent to the investment plans.
- An historic environment with a dwelling community, whose association with the environment might still not be a particularly strong one. This could be the case of Naxos Island, judged by the absence of an active reaction from the local inhabitants to the destruction caused to the historic environment by the transporting of elements of the wind turbines.
- An historic environment with a changing/evolving (not the original) community.
- An historic environment whose community has a claimed special association with the environment. This is the case of the former prisoners of Gyaros Island, who claim a strong association with the place, as illustrated in the establishment of the 'Gyaros Historical Memory' and the organization of ceremonies as a tribute to their departed colleagues, as well as through their firm opposition to the investment plans.

- An historic environment whose community retains its original association with the environment although this association has suffered over time. This could be the case of St George Monastery, which no longer functions as a living monastery in the sense that it does not currently host a monastic community, but is operated by the inhabitants of the local villages on certain occasions every year. It might not be coincidental from this perspective that the inhabitants of the local villages reacted to the investment plans only after the intervention of the NGO and in a rather supplementary and supportive way to the NGO.
- An historic environment whose community retains its original and particularly strong connection with the environment over time to date, and derives a sense of pride, identity, and wellbeing from its connection to the environment. This is the case with Hosios Meletios, which continues to function as a living monastery, hence the monastic community's prompt and firm reaction to the plant's impact on their rituals and life.

Third, REPs see the historic environment, through its association with its local community (in terms of function, material/tangible and intangible expressions, values, and care; see above), as being in continuous evolution over time and into the future, and attempt to associate themselves with this evolution. The current/present phase of a historic environment is not seen as the final one, and thus the emphasis is not only on the REP's main construction phase but also on the pre- and especially the post-construction phase (i.e. the aftermath of the investment).

Fourth, REPs involve the local community in the process from the very beginning, getting to know the characteristics of the community and establishing a formula of cooperation with it. Therefore, REPs consider each historic environment, as defined through its association with its local community, to be unique and in continuous evolution over time to date and into the future, and accordingly approach and involve the local community from the very beginning.

The new approach, as outlined above, could be implemented through the following recommendations (consisting of practical steps) on how an REP project can succeed, while at the same time respecting the historic environment and engaging the local community:

Step 1: Identification and exploration of the historic environment (in relation to the local community)

1. Identification of the historic environment (in relation to its local community).
2. Definition of the boundaries of the historic environment, including:
 - a. the single monument
 - b. the surrounding environment of the monument, and
 - c. the broader landscape.

The boundaries of the historic environment may be defined not only on the basis of the material/tangible expressions but also on the basis of intangible ones. It is important to note that the boundaries of the historic environment should be given increased attention particularly in the case of wind farms because, given that their height can reach and even exceed the 100 m, they can be visible and cause optical interference for a distance of approximately 10 km or even more.¹⁹

3. Definition of the association of the local community with the historic environment, taking into account all the elements that comprise the historic environment: function, material/tangible and intangible expressions, values, and care. Material/tangible and intangible expressions are treated as inseparable, and given equal attention.
4. Taking into account the legislation for the protection of the historic environment.
5. Taking into account the key management systems and the local community. For example, in the case of Hosios Meletios, the Abbot and the Council of the monastic community had to be taken into account from the very beginning of the REP development process.
6. Identification of the existing energy sources, the associated material/tangible and intangible expressions and the (traditional) energy management systems. The existing energy sources might prove to be of interest to the REP, as was learned from the experience at Amorgos Island.

The aim of Step 1 is to determine whether the historic environment in question is suitable for the development of an REP.

Step 2: Cooperating with the local community

1. Identification and exploration of the community. The various communities involved in the operation and management of the historic environment are identified and prioritized (on the basis of their differing associations with the historic environment): the local community; heritage professionals (i.e. state-appointed personnel responsible for the protection of the historic significance of the environment); and the broader community (consisting of all community groups apart from the local community and heritage professionals — such as politicians, researchers, travel agents, and visitors). Emphasis is placed on the local community, which should be studied in terms of its identity, structure, and operation (way of life, decision-making processes, and local resources), and also in terms of its relationship and links with other communities involved in the operation and management of the environment.
2. Establishing a formula of cooperation with the community. The approaching and engaging of the local community should also involve a third person (different from the heritage consultant) who does not have to be a member of the local community — actually, it would be better if they were from outside the local community — but should be aware of the context and the concerns of the local community, and be respected and approved by the community. This person would act as a kind of ‘mediator’ — on the part of the local community — between the community and the REP committee. Also, cooperation with the local community should take a legally binding form, making use of the leaders, the (legal) representatives, and the decision-making process of the community.

The aim of Step 2 is to establish a formula of cooperation with the local community.

Step 3: Mapping of the historic environment (in collaboration with the local community)

1. Mapping of the local community’s present concerns. REPs identify the local community’s concerns about life in general (such as unemployment and human conflict), the historic environment, and the investment.

2. Mapping of the historic environment (through its association with the local community) as it has evolved over time to date, in terms of function, material/tangible and intangible expressions, values, and care (this process is called ‘cultural mapping’). This mapping could help towards identifying the type of historic environment and the strength of the local community’s association with it (see above).
3. Mapping of energy resources and energy management systems over time to date.

The aims of Step 3 are to link the REP development with the protection of heritage and the sustainable development of the local community, and also to link modern energy management systems (of the investor) to traditional ones (of the local community).

Step 4: Assessment and analysis of the historic environment

1. Assessment and analysis of the cultural significance of the historic environment. This involves a hierarchy of heritage expressions and values. For example, in the case of Gyaros Island, the significance of the memory and the associated symbolic value should be given priority over the significance of the material structures and the associated scientific value; in the case of St George Monastery its religious use (by the local community) and its symbolic value (given its role in the Greek Revolution) should be given priority over the historic value of its monastery buildings; and in the case of Hosios Meletios the monastic function should be emphasized.
2. Assessment and analysis of the condition of the (tangible and intangible) expressions of the historic environment.²⁰
3. Assessment and analysis of the broader management context of the historic environment (including legislation, and the power relations between the local community and the heritage professionals and the broader community) that needs to be taken into account at the REP development process.

The aim of Step 4 is to produce a statement about the significance of the historic environment.²¹

Step 5: REP development decisions and actions

1. Considering the impact of the entire REP planning process on the historic environment, with an emphasis on the reversibility of the actions taken. The pre- and post-construction phases are considered to be equally as important as the main construction phase. Specifically, the pre-construction phase may require actions such as the deforestation of an extended area for ensuring there is sufficient space for the construction of the REP, and the opening of a road network for transferring the appropriate tools and structures,²² as well as the construction of deep foundations.²³ These actions may result in a serious impact on the historic (as well as the natural) environment, causing a downgrade of the social and economic development of the local community, and even the mobility of communities and the appearance of ‘environmental migrants/refugees’.²⁴ As far as the post-construction phase is concerned, REPs are not seen as additions of certain structures (e.g. water dams or wind

turbines) to an existing historic environment; in fact, they may constitute the next phase in the evolution of the historic environment (through its association with the local community, seen in terms of function, material/tangible and intangible expressions, values, and care) over time and into the future. It is important to stress that this new phase in the evolution of the historic environment is by no means the final one, and will be followed by other phases in the future.

2. Identifying and communicating the benefits brought by the REP development to the local community, in other words, the possible ways in which the investment could fit within and serve the community's concerns (as described above).
3. Identifying specific ways to involve the local community in the process, by mobilizing and utilizing local resources (such as human and material resources). This could help in creating a stronger link between the local community and the investment — a sense of 'belonging'.

The aim of Step 5 is to prepare an action/development plan.

Step 6: Periodic evaluation/review and revision of decisions

The aim of Step 6 is to evaluate and revise the development of REPs on the basis of the protection of the historic environment and the sustainable development of the local community.

It is important to note that these recommendations should be applied through a heritage consultant.²⁵ This person should be involved from the very beginning of the REP development process (Step 1) and, depending on the significance of the historic environment and the concerns of the local community in question (see above), may be an external consultant or a member of the REP committee for the planning of the investment. This person should be in close collaboration — acting on behalf of the REP committee — with the 'mediator', who acts on behalf of the local community (see above).

Concluding remarks

Cultural and social aspects are of crucial importance for the development of REPs, particularly in the historic environment. Therefore, a new approach is suggested which links the development of REPs to the protection of the historic environment and the sustainable development of the local community. This strategy attempts to point towards a different way of understanding, planning, and implementing the development of REPs. A REP should not be seen only as an external intervention to a historic environment, with — almost by definition — a rather negative impact on it; rather, the focus should be on how to limit this impact for the sake of the protection of the historic environment. A REP may now be seen also as part of the historic environment, in accordance with the specific characteristics of it (as defined through its association with the local community), as another step — not the final one — in the evolution of the historic environment over time and into the future. Accordingly, our focus should be on how to manage, support, and guide this evolution.

Acknowledgements

This paper benefits from my experience as a heritage consultant with an emphasis on community issues with the ICCROM *Living Heritage Sites* programme under the supervision of Gamini Wijesuriya and Joe King, to whom I am grateful. This work benefits from the work of other colleagues carried out throughout the development of the programme, and I would like to thank ICCROM for allowing me access to this work and also for approving its dissemination. I owe special thanks to Nicholas Stanley-Price and Giovanni Cafiero for most interesting discussions on the impact of the development of REPs on historic environments. I would also like to thank Dora Konsola, Stelios Lekakis, Eirini Gratsia, and Maria Scalia for providing me with most useful material concerning the case studies discussed in this article. Thanks also to Eirini Gratsia and Giorgos Chondros for providing the photographs used in the text.

Earlier forms of this research were presented at the Conference on Green Development and the Energy Planning of Western Greece organized by the Institute for the Development of Western Greece (INADE) in Agrinio, Greece, 27 November 2010, and at the International Conference on Energy Management in Cultural Heritage, organized by UNDP Croatia and UNESCO Venice Office in Dubrovnik, Croatia, 6–8 April 2011.²⁶ I take this occasion to thank the organizing bodies of the conferences for their hospitality and the participants for their comments. Parts of this research were presented at the UNESCO Regional Schools on Renewable Energy and Energy Efficiency Governance, organized by the UNESCO Venice Office in collaboration with local and regional organizations in Fojnica, Bosnia and Herzegovina, 23 August–3 September 2011, and in Dubrovnik, Croatia, 30 September–6 October 2012 where I was invited to participate upon the initiative of Davide Poletto, whom I thank warmly.

Notes

¹ G. Papandreou, Μια κριτική προσέγγιση για τις ΑΠΕ (2011) <<http://www.papandreou.gr/papandreou/content/Document.aspx?m=12892&rm=16718289&l=2>> [accessed 20 February 2011].

² Greek Government 2001 Law 2941/2001 «Απλοποίηση διαδικασιών ίδρυσης εταιρειών, αδειοδότησης Ανανεώσιμων Πηγών Ενέργειας, ρύθμιση θεμάτων της Α.Ε. «ΕΛΛΗΝΙΚΑ ΝΑΥΠΗΓΕΙΑ» και άλλες διατάξεις», Government Gazette 201/A/12-9-2001, article 2.

³ Greek Government 2002 Law 3028/2002 «για την προστασία των Αρχαιοτήτων και εν γένει της Πολιτιστικής Κληρονομιάς», Government Gazette 153/A/28-6-2002, article 3; Council of Europe 1985 Convention for the Protection of the Architectural Heritage of Europe <<http://conventions.coe.int/treaty/en/treaties/html/121.htm>> [accessed 20 January 2011]; Council of Europe 2000 European Landscape Convention <http://www.coe.int/t/dg4/cultureheritage/heritage/landscape/default_en.asp> [accessed 20 January 2011]; see also A. Papakonstantinou, Η Σύμβαση της

Γρανάδας για την Προστασία της Αρχιτεκτονικής Κληρονομιάς και το Σύνταγμα. Με αφορμή της αποφάσεις ΣτΕ 2300/1997 (Ολ) και 3113/1998, [The Granada Convention for the Protection of the Architectural Heritage of Europe and the Greek Constitution] Journal Νόμος και Φύση (March 1999) <<http://www.nomosphysis.org.gr/articles.php?artid=27&lang=1&catid=1>> [accessed 24 November 2010]; A. Papakonstantinou, Το Νομικό Καθεστώς των Ανανεώσιμων Πηγών Ηλεκτρικής Ενέργειας, Journal Νόμος και Φύση (July 2004) <<http://www.nomosphysis.org.gr/articles.php?artid=248&lang=1&catid=1>> [accessed 4 November 2010]; P. Patronos, V. Karageorgiou, and A. Papapetropoulos, Διεθνείς και Κοινοτικές Δεσμεύσεις της Ελλάδας για την Προώθηση των Ανανεώσιμων Πηγών Ενέργειας (Γνωμοδότηση), Journal Νόμος και Φύση (July 2004) <<http://www.nomosphysis.org.gr/articles.php?artid=244&lang=1&catid=1>> [accessed at 24 November 2010].

- ⁴ Greek Government 1975 Constitution of the Hellenic Republic, article 24; also Greek Government 2002.
- ⁵ Greek Government 2002, article 40.
- ⁶ Greek Government 2002, article 1.
- ⁷ For a critique of the system of heritage protection in Greece see K. Kotsakis, 'The Powerful Past: Theoretical Trends in Greek Archaeology', in *Archaeological Theory in Europe: The Last Three Decades*, ed. by Ian Hodder (London: Routledge, 1991), pp. 65–90; Ioannis Poullos, Αρχαιολογική Προστασία και Μοναστική Χρήση: η «Ζώσα» Μονή Παντάνασσας μέσα στη «Νεκρόπολη» του Μυστρά, [Archaeological Protection and Monastic Use: the 'Living' Pantanassa Monastery within the 'Necropolis' of Mystras], *Journal Βυζαντικά*, 29 (2011), 353–69 (p. 353–70; abstract in English p. 369); Ioannis Poullos, Εισαγωγή στη Διαχείριση της Πολιτισμικής Κληρονομιάς: Ορολογία, Έννοια και Μοντέλα Διαχείρισης, [An Introduction to Cultural Heritage Conservation/Management: Terminology, Meaning, and Approaches] *Journal Αρχαιολογία* (in press, 2012).
- ⁸ Ministry of Culture 2003 Presidential Decree 191, «περί Οργανισμού Υπουργείου Πολιτισμού», *Government Gazette* 146/A/13-6-2003; Greek Government 2002, articles 49–50.
- ⁹ For a more detailed account of the difficulties see: D. Konsola, 'Αναπτυξιακά Έργα και Πολιτιστική Κληρονομιά: Η Περίπτωση των Έργων Εκτροπής του Αχελώου', in Στο Βέλος του Χρόνου (Τιμητικός τόμος για τον ακαδημαϊκό-καθηγητή Γεώργιο Π. Λάββα) (Thessaloniki: University Studio Press, 2004), pp. 365–73.
- ¹⁰ S. Garozi, Σχολιασμός Υπόθεσης Αχελώου (ΣτΕ 2760/1994 και 3478/2000) <<http://www.jurisconsultus.gr/pubs/uploads/1540.pdf>> [accessed 20 January 2011]; Konsola 2004; Technical Chamber of Greece 2005 Proceedings of the Conference on 'Αχελώος: Η Βέλτιστη Διαχείριση', Technical Chamber of Greece, Athens, 1–2 December 2005, Athens Technical Chamber of Greece <http://library.tee.gr/digital/m2099/m2099_me_energias.pdf> [accessed at 24 November 2010]; see also Ioannis Poullos, 'Energy Investments, Historic Environments and Local Communities: Lessons from the Greek Experience', *Proceedings, International Conference on Energy Management in Cultural Heritage* (UNDP Croatia and UNESCO Venice Office), Dubrovnik, Croatia, 6–8 April 2011 <http://www.an-patrimoine-echanges.org/IMG/pdf/session_s4-ioannis_poullos_-_universite_de_grece_-_athenes_grece.pdf> [accessed 2 October 2011].
- ¹¹ P. Giogiakas, «Οι ανεμογεννήτριες βλάπτουν το σεξ»! Ιστορίες αιολικής τρέλας από όσους αντιδρούν στην εγκατάστασή τους στην Ελλάδα (2008) <<http://www.tanea.gr/default.asp?pid=2&ct=1&artid=4488429>> [accessed 25 November 2010]; Stelios Lekakis, personal communication: 'Monumenta' magazine for the protection of natural and architectural heritage in Greece and Cyprus, and ELLINIKI ETAIRIA Society for the Environment and Cultural Heritage; Eirini Gratsia personal communication: 'Monumenta' magazine for the protection of natural and architectural heritage in Greece and Cyprus, and ELLINIKI ETAIRIA Society for the Environment and Cultural Heritage.
- ¹² D. Myrilla, Κόκκινη Γυάρος, Πράσινη Ανάπτυξη, Newspaper Kyriakatiki Eleftherotypia, 17 October 2010 <<http://www.enet.gr/?i=issue.el.home&date=17/10/2010&cid=213887>> [accessed 22 January 2011]; N. Kontraru-Rassia, Αρχαία και Πράσινη Ενέργεια, Newspaper Κυριακάτικη Ελευθεροτυπία, 28 May 2011; M. Thermou, Νέα Όρια του Ιστορικού Τόπου της Γυάρου, Newspaper Το Βήμα, 13 May 2011 <<http://www.tovima.gr/culture/article/?aid=400634>> [accessed 13 May 2011]; M. Mani, Γυάρος — Από τόπος εξορίας, 'φιλέτο', Newspaper Κεφάλαιο, 28 May 2011 <<http://www.capital.gr/news.asp?id=1205426>> [accessed 28 May 2011]; Ch. Floudopoulos, Μεγαλόπνοια projects 'παγωμένα' από το 2005, republished from newspaper Κεφάλαιο, 21 May 2011 <<http://www.capital.gr/news.asp?id=1205426>> [accessed 28 May 2011]; Monumenta, Δικαιοσύνη για τον Αχελώο!, 11 February 2010 <<http://www.monumenta.org/article.php?perm=1&IssueID=2&Lang=gr&CategoryID=20&ArticleID=473>> [accessed 2 May 2011].
- ¹³ Naxos Blog, Αιολικό Πάρκο στην περιοχή της Κωμιακής (2008) <<http://apollonasnaxou.blogspot.com/2008/08/blog-post.html>> [accessed 20 January 2011]; Stelios Lekakis, personal communication.
- ¹⁴ G. Kiouisis, Αμоргός: Διαδρομές νερού, Newspaper Ελευθεροτυπία, 18 November 2009 <<http://www.enet.gr/?i=news.el.article&cid=103099>> [accessed 21 January 2011].
- ¹⁵ Ioannis Poullos, 'Is Every Heritage Site a "Living" One?: Linking Conservation to Communities' Association with Sites', *The Historic Environment: Policy and Practice*, 2.2 (2011), 144–56; Ioannis Poullos, 'Moving Beyond a "Values-Based Approach" to Heritage Conservation', *Conservation and Management of Archaeological Sites*, 12.2 (May 2010), 170–85 (pp. 175–76).
- ¹⁶ R. Mason, 'Assessing Values in Conservation Planning: Methodological Issues and Choices', in *Assessing the Values of Cultural Heritage: Research Report*, ed. by M. de la Torre (Los Angeles: The Getty Conservation Institute, 2002), pp. 5–30 (p. 27); R. Mason and E. Avrami, 'Heritage Values and Challenges of Conservation Planning', in *Management Planning for Archaeological Sites*, ed. by J. M. Teutonico and G. Palumbo, International Workshop organized by the Getty Conservation Institute and Loyola Marymount University, May 2000 (Los Angeles: The Getty Conservation Institute), pp. 13–26 (p. 15).

- ¹⁷ E. Edroma, 'The Notion of Integrity for Natural Properties and Cultural Landscapes', in *Authenticity and Integrity in an African Context*, ed. by G. Saouma-Forero, Expert Meeting, Great Zimbabwe National Monument, Zimbabwe, 26–29 May 2000 (Paris: UNESCO, 2001), pp. 50–58 (pp. 55–56); W. Ndoro, 'Traditional and Customary Traditional Systems: Nostalgia or Reality? The Implications of Managing Heritage Sites in Africa', in UNESCO Netherlands, *Linking Universal and Local Values: Managing a Sustainable Future for World Heritage*, conference organized by the Netherlands National Commission for UNESCO, in collaboration with the Netherlands Ministry of Education, Culture, and Science (22–24 May 2003) (Paris: UNESCO World Heritage Centre, 2004), pp. 81–84 (p. 81).
- ¹⁸ Poulos, 'Is Every Heritage Site a "Living" One?'; Ioannis Poulos, 'Conserving Living Religious Heritage: Maintaining Continuity and Embracing Change — The Hindu Temples of Tanjore, Srirangam and Tirupati in India' (South Asian Arts, October 2010) <www.southasianarts.org> [accessed 1 November 2010].
- ¹⁹ English Heritage, *Wind Energy and the Historic Environment* (London: English Heritage, 2005) <http://www.helm.org.uk/upload/pdf/Wind_Energy_%28final29.pdf> [accessed 24 November 2010], pp. 4–7.
- ²⁰ For further details see: M. Demas, 'Planning for Conservation and Management of Archaeological Sites', in *Management Planning for Archaeological Sites*, ed. by J. M. Teutonico and G. Palumbo, International Workshop organized by the Getty Conservation Institute and Loyola Marymount University, May 2000 (Los Angeles: The Getty Conservation Institute, 2002), pp. 27–54.
- ²¹ Demas.
- ²² English Heritage, *Strategic Environmental Assessment, Sustainability Appraisal and The Historic Environment* (London: English Heritage, 2009) <<http://www.helm.org.uk/upload/pdf/Strat-env-ass.pdf?1307719753>> [accessed 24 November 2010], p. 10.
- ²³ As is the case with onshore and offshore wind farms, see: English Heritage, *Wind Energy and the Historic Environment* (London: English Heritage, 2005) <http://www.helm.org.uk/upload/pdf/Wind_Energy_%28final29.pdf> [accessed 24 November 2010], pp. 4, 10–11; COWRIE, *Historic Environment Guidance for the Offshore Renewable Energy Sector* (prepared by Wessex Archaeology Ltd, 2007) <http://www.offshorewind.co.uk/Assets/archaeo_guidance.pdf> [accessed 24 November 2010]; COWRIE, *Guidance for Assessment of Cumulative Impacts on the Historic Environment from Offshore Renewable Energy* (prepared by Oxford Archaeology with George Lambrick Archaeology and Heritage, 2008) <http://www.offshorewind.co.uk/Assets/cowrie_ciarch%20web.pdf> [accessed 24 November 2010].
- ²⁴ L. Brown, P. McGrath, and B. Stokes, 'Twenty-Two Dimensions of the Population Problem', Worldwatch Paper 5 (Washington, DC: Worldwatch Institute, 1976); International Organisation for Migration, *Discussion Note: Migration and the Environment*, 94th session, MC/INF/288, 1 November 2007 <http://www.iom.int/jahia/webdav/shared/shared/mainsite/about_iom/en/council/94/MC_INF_288.pdf> [accessed 2 December 2011]; A. Katsaris, Κλιματική Αλλαγή και Περιβαλλοντικοί Πρόσφυγες, *Journal Νόμος και Φύση* (May 2010); see also G. Tsaltas, Επί Ξηρού Ακμής, in *Περιβάλλον και Πολιτισμός. Προβληματισμοί από το Παρελθόν. Προτάσεις για το Μέλλον*, Ελληνική Εταιρεία Περιβάλλοντος και Πολιτισμού — Ωκεανίδα, ELLINIKI ETAIRIA Society for the Environment and Cultural Heritage — Oceanida ed. by F. Νεοφύτου (Athens, 2010), pp. 169–75 (p. 173).
- ²⁵ English Heritage, *Strategic Environmental Assessment*, p. 10.
- ²⁶ Poulos, 'Energy Investments'.

Notes on contributor

Ioannis Poulos undertook his PhD on heritage conservation and management at University College London, and attended MBA courses on business strategy and management at London Business School. Ioannis has collaborated as a heritage consultant with ICCROM and with local Greek heritage organizations, and is a lecturer at the University of Western Greece and at the Hellenic Open University.

Correspondence to: Ioannis Poulos, 40 Peloponnissou Street, Agia Paraskevi, 15341 Athens, Greece. Email: jannispoulos@hotmail.com